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Task Description:

NOTE 1/21/2020: Continuation of "Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids," grant NNX13AQ53G, with same Principal Investigator Dr. Boris Khusid.
Motivation: The widespread use of colloidal processes for scalable manufacturing of structured materials emphasizes a critical need for improving fundamental understanding of the role of external fields in directing non-equilibrium phenomena in suspensions. The challenge is due to kinetic limitations because the particles can be trapped into metastable configurations for a long time due to the lower mobility of multi-particle structures compared to that of individual particles. Microgravity offers a unique opportunity to study these phenomena by removing masking gravity effects, such as particle sedimentation, convection, and jamming. The proposed research addresses both fundamental and technological questions in the science of colloids aimed at understanding the equilibrium and metastable crystalline, liquid, and glassy structures and the use of these materials in additive manufacturing.

Objectives: Conduct tests in the International Space Station (ISS) Advanced Colloids Experiment (ACE) facility to elucidate the mechanisms of non-equilibrium phenomena underlying the assembly of colloidal particles assisted by temperature field gradients and suggest novel routes for processing functional materials.

Methodology: A novel approach will be used to study mechanisms for formation of metastable and glassy phases in suspensions in the ISS and for comparison on Earth. A single sample will be exposed to a temperature gradient to cover the interesting range of particle densities. As the particle density is directly measured by microscopy, a priori knowledge of the gradient profile is not required. Experiments will involve setting up a temperature gradient to observe the resulting structures and then locally mix a region of known density to watch it glassify or crystallize. Quantitative data on the suspension rheology will come from microrheology measurements through tracking particle thermal motion.

Deliverables: Understanding of non-equilibrium phenomena in colloids driven by temperature gradients and experimental database for the control and manipulation of colloidal structures in space and terrestrial applications.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Research Overview
• Why is the research needed? New functional materials are created using micron-sized particles suspended in fluid (called colloids) that self-organize into crystalline structures or amorphous glass phases by means of entropic forces or under the control of non-equilibrium drive as supplied, for example, by temperature gradients. The ACE-T11 experiments in the ISS utilizes confocal 100 X microscopy for time- and space-resolved 3D imaging of the arrangement of spherical colloidal particles, and examines the influence of temperature gradients on the particle motion and arrangement (referred to as thermophoresis).

• What is accomplished? In ACE-T11, the phase behavior of micrometer-sized colloidal particles in long-duration microgravity is studied in the ISS on dense suspensions at volume fractions f~0.60-0.63. The particles are found to self-organize into face centered cubic (FCC) colloidal crystal of the capillary size (27 mm x 1.5 mm x 150 μm). It is the first confirmation of the theory for the phase behavior of hard spheres, the simplest model of matter with a crystallization transition. In contrast, colloidal crystals formed on Earth do not grow larger than several hundreds of micrometers due to gravitational settling.

• What is the impact of the research? Ultimately, the ability to design functional structures – based on micrometer-scale building blocks – with a variety of well-controlled three-dimensional bonding symmetries, amorphous structures and different rheologies will allow the development of new devices for chemical energy production and storage, photonics and communication, and a new set of slurries and pastes useful for additive manufacturing. Such materials might include photonic crystals with programmed distributions of defects. Optical technology utilizing such materials may offer intriguing solutions to unavoidable heat generation and bandwidth limitations facing the computer industry. New insights were gained in the ISS experiments on the formation of crystalline phases as distinct from the amorphous glass, a question raised by previous microgravity studies, as yet unresolved, whether glass phases found on Earth would readily crystallize in microgravity. In particular, it was revealed that molecular dynamics simulations of equilibrium work when gravitational effects are unimportant. They therefore can be used for simulations of materials processing in microgravity.

• Space Applications: Eventually, future space exploration may use self-assembly and self-replication to make materials and devices that can repair themselves. Self-assembly and evolutionarily-optimized functional units are key to long-duration space voyages. Even more immediate is the requirement of replacement parts and specialized repair facilities for space missions. 3D printing and additive manufacturing will be necessary for future space missions. The development of particle slurries and pastes with the appropriate rheological properties that work in both microgravity and conventional gravity will be needed. One objective of the experiments conducted in the ISS is to develop new materials that cannot be formed on Earth.

• Earth Applications: This investigation involves several fundamental and practical aspects of soft matter science with potential applications on Earth. Self-assembly processes are crucial to making functional materials and devices from small particles. Improved design and assembly of structures fabricated in microgravity may have use in a variety of fields from medicine to optoelectronics on Earth. Ultimately, the ability to design and build functional structures based on colloids will allow new devices for chemical energy, communication, and photonics, including photonic materials to control and manipulate light. The rapidly growing fields of 3D printing and additive manufacturing rely on the assembly and sintering of particle aggregates and the preparation of high-density slurries and pastes of different colloidal materials and with different rheological and mechanical properties is a main goal of these studies.

Motivation:
Packing of hard spheres has fascinated mankind for millennia before it has been recognized to also serve as a versatile model for many phenomena in atomic solids. The densest packing of spheres at the volume fraction $\varphi=\pi/\sqrt{18} \sim 0.74$, conjectured by Kepler in 1611 and proven by Hales in 1994 is considerably higher than disordered packing with $\varphi<0.64$ found so far. Face-centered cubic (FCC) and hexagonal close-packed (HCP) are both close-packed lattices of spheres that differ in how their layers of spheres arranged at the vertices of a triangular tiling are stacked. The spheres in the first and third layers lie directly above each other in HCP. In FCC, each layer is slightly shifted relative to the previous one, resulting in different alignment between the first and third layers. A difference in packing fraction between the crystals and the amorphous structures forms the basis of much of what we know about the transition from liquid to crystal in the

periodic table of elements.

Hard spheres are impenetrable spheres that cannot overlap in space. They are used as a model to describe the interactions between atoms and molecules. The entropy-driven phase transition in a hard-sphere system from a liquid to a crystal is a paradigm success story of statistical mechanics that shows the utility of computer simulations in discovering the properties of matter. An “entropy-driven phase transition" refers to a phase change in a system where the driving force is the increase in overall disorder or entropy, meaning that the system transitions to a state with higher microscopic randomness even if it appears more ordered on a macroscopic level. Theories in the 1930’s-60's predicted that a hard-sphere system forms a liquid phase for a particle volume fraction up to φ=0.494, a FCC phase above φ =0.545, and coexistence in between. In the mid 1980's, Pusey and van Megan confirmed this phase transition in colloids of closely matched particle/solvent densities. However, their work and numerous experiments on Earth that followed, observed small sedimented crystallites of mixed FCC and HCP structures with many stacking faults and a high-density glass phase, in disagreement with predictions.

Early microgravity experiments in the Space Shuttle on hard-sphere colloids demonstrated the absence of the glassy phase. No evidence of glass transition was also observed later in large-scale molecular dynamics simulations of monodispersed hard spheres performed for sufficiently long equilibration times. We showed that suppression of crystallization of a metastable liquid-phase in high-density colloids observed in terrestrial experiments cannot be attributed to cessation of long-time particle diffusivity, as a growing crystallite is surrounded by a depleted colloidal fluid. The one important remaining question is which of the stackings/packings has the highest entropy and forms the lowest free energy, hence stable, crystal. Simulations predict that FCC has a higher entropy by ~0.001 per sphere than HCP stackings and slightly lower surface tension. The picture from simulations for monodisperse hard spheres in equilibrium is that there is a low-density fluid phase, a coexistence region, and a high density FCC crystal with no glass phase. Our experiment on the International Space Station (ISS) was designed to answer the question of the crystal structure of the stable, lowest free energy configuration of hard spheres.

ISS colloidal experiments:

Our refractive-index colloids are similar to those used by Pusey and van Megan: poly(methyl methacrylate) spheres coated with poly(12-hydroxystearic acid), dry diameter 1.25±0.05 μm with polydispersity 4.3% dispersed in a decalin/tetralin mixture. The rhodol julolidine fluorophore molecules are covalently linked inside the Poly(methyl methacrylate) (PMMA) spheres for observation in an automated confocal microscope of the ISS Light Microscopy Module (LMM). Previous microgravity and Earth based samples were mounted in cm scale cuvettes. The present samples, referred to as C1, C2, and C3 were filled and sealed on the Earth in three rectangular glass capillaries, 50mm x 2mm x 0.2mm, mounted in a temperature-controlled Sample Module with the observation area, 40mm x 1.6mm x 0.15mm, and then delivered to the ISS. All samples were studied in two sessions respectively after 9 and 17 months from the delivery to the ISS, each time over 2 months. Since the LMM was not equipped with conventional light scattering technique, a slow scanning mode of confocal microscopy in the plane parallel to the capillary cover and a stacking mode perpendicular to the cover were used to study the colloidal structures. Images were acquired with a 100x oil immersion objective with the image size 109.44 x 80.40 μm in the plane parallel to the capillary cover. 130 GB of confocal images were collected and processed with the open-source ImageJ 2.3.0 and MATLAB R2022a. No statistically significant difference in images of colloidal structures collected in the first and second sessions was noted.

Task Progress:

Colloidal structure in sample C3:

Our ISS experiments demonstrate that exposure to microgravity formed highly homogeneous particle distribution. Remarkably, the stitched confocal images reveal no dislocations, grain boundaries, and topological defects away from the capillary side walls by ~ 0.4 mm. These observations indicate that a 3D crystal formed in the observable region of the sample capillary away from the walls and the single bubble.

Images at different locations are taken to compute the two-dimensional radial distribution functions (or pair correlation functions) to verify the particle ordering Averaging data over all images yields 1.32±0.02 μm for the particle diameter (5.6% larger than the dry size due to swelling) and 1.38±0.02μm for the separation between particle centers. The computed planar density of this hexagonal lattice, 0.61±0.02 1/μm^2, is in excellent agreement with measurements.

To address the real question whether we have a coherent crystal, we stitch together images taken at different locations away from the capillary cover and draw straight lines through particle centers in contiguous images. The longest line is over 120 stitched images, ~ 5,000 interparticle spacings. We then take a one-dimensional discrete Fourier Transform of the greyscale through these lines. All of them demonstrate two sharp peaks whose peak heights increase linearly with the number of particles (~30 per image) along this line. Incoherence from grain boundaries would have broadened, not just damped, the peaks. Calculations indicate that a linear increase of the peak heights with the number of particles occurs only in the case of translational periodicity of the particle center locations. The ratio between heights of the second and first peaks for all scans randomly fluctuates around 0.3 from image to image. For this ratio, the expression for the one-dimensional Debye-Waller factor yields ~ 0.20 μm for the mean square displacement of the particle centers from lattice nodes. Averaged over images taken far away from one another, the ratio of peaks between the second and first sextets of the Bragg spots in two-dimensional discrete Fourier Transform spectra is also ~0.3. For this ratio, the expression for the two-dimensional Debye-Waller factor yields ~ 0.19 μm. Both values are close to 0.15±0.09 μm measured by tracking in videos the particle Brownian motion. Remarkably, the ratio of the root-mean-square displacement of particles away from the lattice node and the distance between them, 0.11-0.14, slightly exceeds the famous Lindemann melting criterion ~0.1 for melting to occur, that is well observed for mono-atomic solids. This fact indicates very intensive Brownian motion of particles around the crystal nodes. To find the crystal structure, we identify the order of the stacking planes. Three distances calculated from plane spacing and in-plane distances corroborate the FCC structure. As the size of the FCC lattice constant is now confirmed, we can compute the volume fraction of particles in the crystal φ=0.65±0.04, consistent with the initial volume fraction and moderate swelling of the particles.

Colloidal structures in sample C1 and C2:

Sample C1 dried out and lost transparency in the bulk. Images taken in the middle of the sample under the capillary cover show a thin layer of irregular patches of dried particles arranged in a well-defined hexagonal lattice. The wet particles seen near these patches, as well as under them, appear to be disordered. Interestingly, similar 2D honeycomb structures were observed in previous ISS experiments on polymer-based colloids.

The colloidal structure formed in sample C2 exhibits both crystalline and non-crystalline regions. However, the area of

	crystalline domains reduces rapidly with increasing their distance from the capillary cover. The reason is that particle ordering away from the capillary cover is suppressed by displacement and shear generated in the capillary. Discussion and outlook: Formation of a large FCC colloidal crystal with the particle volume fraction φ≈0.65 in the ISS confirms predictions for the equilibrium phase diagram of hard spheres, the conclusion of theoretical, simulation, and experimental studies over at least the past 80 years. It also confirms the accuracy of molecular dynamics simulations to predict the equilibrium structure of a large crystal, even when the free energy of nearby states differs by only one part in a thousand. The discovery that the absence of gravity makes it possible to equilibrate colloids suggests a new route to fabricate colloidal crystals for mid-infrared photonics in a low Earth orbit (LEO) whose grating constant can be varied by changing the particle size and volume fraction. Sample C3 sits in our lab. Reentry or the Earth gravity rendered it into ~0.3 mm crystallites with clear cracks and other defects. However, we could solidify/crosslink the solvent to fix the crystal in LEO and return it to Earth.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Abstracts for Journals and Proceedings	Lei Q, Khusid B, Hollingsworth AD, Chaikin PM, Meyer WV, Ibrahim A, Jacob Scoccimerra J. "Fabrication of 3D colloidal crystals for infrared photonics in low-Earth orbit (LEO)." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Articles in Peer-reviewed Journals	Boymelgreen A, Schiffbauer J, Khusid B, Yossifon G. "Synthetic electrically driven colloids: A platform for understanding collective behavior in soft matter." Current Opinion in Colloid & Interface Science. 2022 Aug 1;60:101603. https://doi.org/10.1016/j.cocis.2022.101603 , Aug-2022